🏢 **Visitor Management System – J.Ragumaan**

Project Documentation

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1. Introduction

The **Visitor Management System (VMS)** is a web-based application designed to manage and track visitors entering an organization.
The system replaces manual entry registers with a digital platform that allows visitors to book slots in advance and enables administrators to manage and approve visitor requests efficiently.

The application is built using **React** for the frontend, **Spring Boot** for the backend, and **MySQL** as the database.

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2. Problem Statement

Traditional visitor management using paper registers has several drawbacks:

- No real-time visitor tracking
- Lack of approval workflow
- Data inconsistency and loss
- No visibility of visitor status
- Time-consuming manual verification

The objective of this project is to build a **secure, user-friendly, and efficient visitor management system** that automates the visitor entry process.

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3. Objectives

- To allow visitors to book visit slots online
- To track visitor visit status (Pending, Approved, Completed)
- To provide an admin dashboard for approval and management
- To reduce manual work and improve security
- To store visitor data digitally for future reference

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4. Scope of the Project

In Scope

- Visitor slot booking
- Visitor status checking
- Admin approval and deletion of visitors
- Status-based workflow
- Responsive user interface

Out of Scope

- Authentication using OTP or biometrics
- Multiple admin roles
- Visitor photo capture
- Entry gate hardware integration

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5. Technology Stack

Layer	Technology
Frontend	React, Bootstrap, Font Awesome
Backend	Spring Boot, Spring Data JPA
Database	MySQL
ORM	Hibernate
Build Tool	Maven
Version Control	Git, GitHub

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6. System Architecture

The system follows a **three-tier architecture**:

1️⃣ Frontend (React)

- Visitor booking form
- Visitor status checking screen

- Admin dashboard for visitor management
- Communicates with backend via REST APIs

2️⃣ **Backend (Spring Boot)**

- Handles business logic
- Validates visitor data
- Manages visitor status workflow
- Communicates with database

3️⃣ **Database (MySQL)**

- Stores visitor details and status
- Maintains visit history

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7. Functional Modules

7.1 Visitor Module

- Enter visitor details
- Select purpose of visit
- Choose person to meet
- Receive booking confirmation

7.2 Visitor Status Module

- Check visit status using email or ph no
- Display visitor name, purpose, meet person
- Show current visit status

7.3 Admin Module

- View all visitor records
- Approve pending visitors
- Mark visits as completed after business hours
- Delete visitor records

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8. Database Design

Table: visitor

Column Name	Data Type	Description
id	BIGINT (PK)	Unique visitor ID
name	VARCHAR	Visitor name
email	VARCHAR	Visitor email
phone	VARCHAR	Contact number
meet_person	VARCHAR	Person to meet
purpose	VARCHAR	Purpose of visit
status	ENUM	PENDING / APPROVED / COMPLETED
created_at	TIMESTAMP	Visit request time

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9. ER Diagram (Description)

The system uses a **single entity Visitor**, which stores all visitor-related information. The visitor status attribute controls the workflow from booking to approval and completion.

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10. Backend Design

REST APIs

Method	Endpoint	Description
POST	/api/visitors	Create visitor booking
GET	/api/visitors	Fetch all visitors (Admin)
GET	/api/visitors/status	Get visitor status by email
PUT	/api/visitors/{id}/approve	Approve visitor
DELETE	/api/visitors/{id}	Delete visitor

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11. Visitor Status Workflow

PENDING → APPROVED → COMPLETED

- **PENDING**: Visitor has booked a slot
- **APPROVED**: Admin has approved the visit
- **COMPLETED**: Visit is finished

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12. Frontend Design

UI Screens

- Welcome screen (Visitor / Visitor Status / Admin)
- Visitor booking modal
- Booking success confirmation
- Visitor status check screen
- Admin dashboard with visitor table

UI Features

- Modal-based forms
- Status badges with color coding
- Responsive layout
- Clean and minimal design

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13. Application Flow

1. Visitor opens application
2. Visitor fills booking form
3. Backend stores visitor with status **PENDING**
4. Admin views visitor in dashboard
5. Admin approves visitor
6. Visitor checks status
7. Status updates to **APPROVED / COMPLETED**

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14. Validations and Edge Cases

- Mandatory field validation
- Email format validation
- Duplicate visitor handling
- Status cannot skip workflow
- Only pending visitors can be approved
- Deleted visitors removed from active list

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15. Advantages of the System

- Digital visitor tracking
- Improved security and control
- Faster approval process
- User-friendly interface
- Reduced manual errors

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16. Limitations

- No login authentication
- No real-time notifications
- Single admin role
- No visit rescheduling

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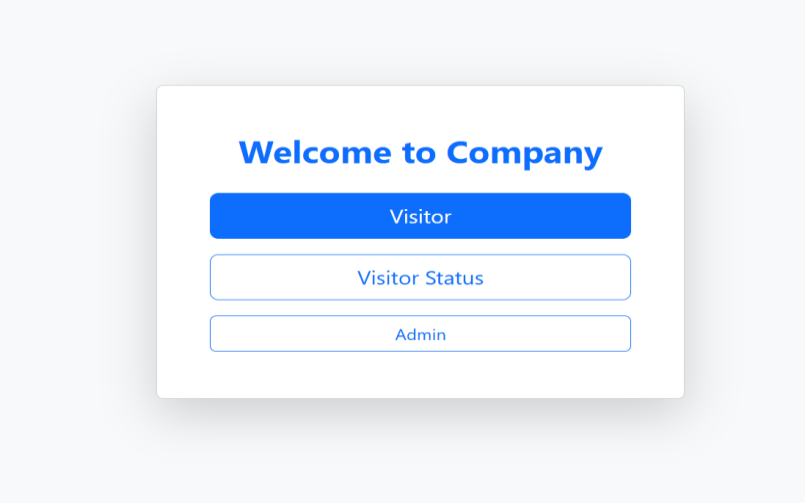
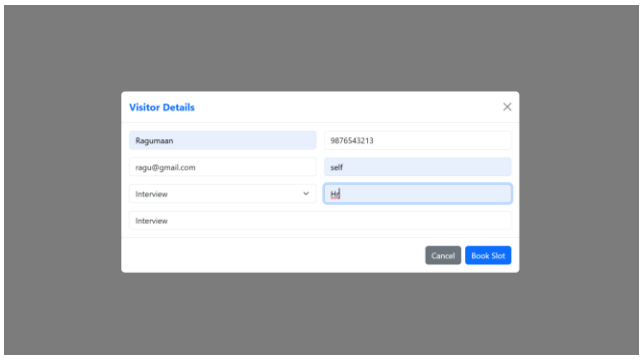
17. Future Enhancements

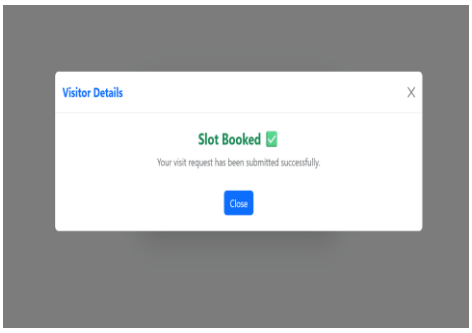
- Email or SMS notifications
- Visitor QR code generation
- Multi-admin support
- Entry/exit time tracking
- Visitor photo upload

18.Screenshot

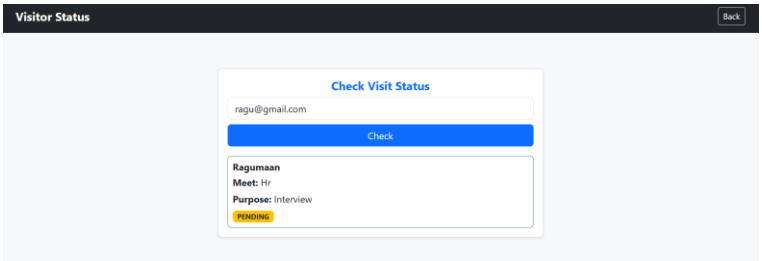


Landing page

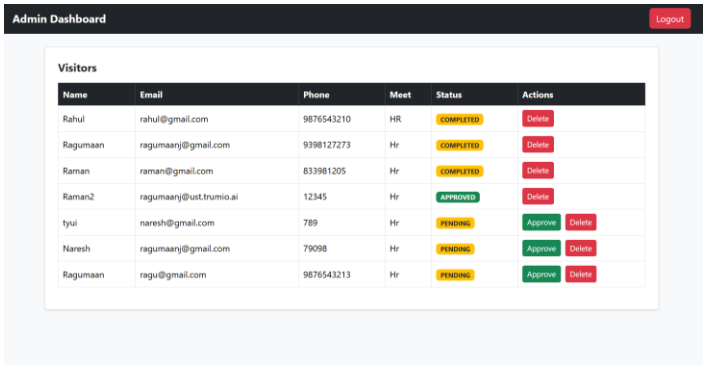




Visitor Form and slot Booked details

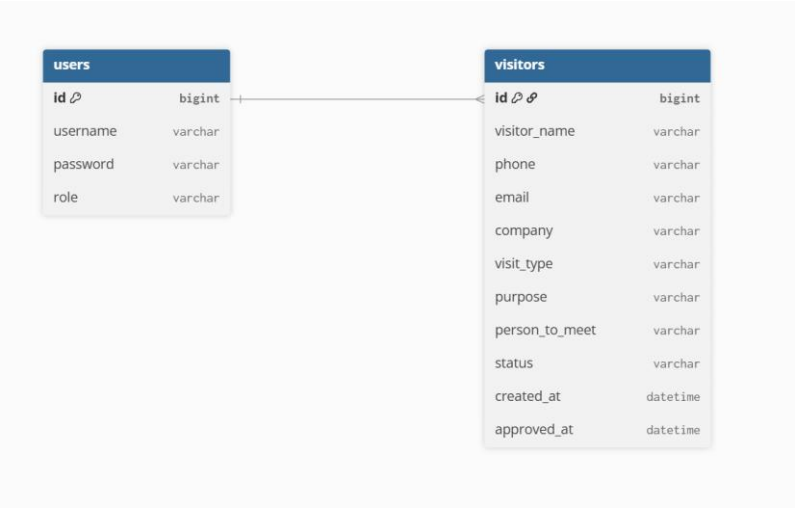


Check Visitor status



Admin dashboard

19.ER Diagram



User=admin and Visitor = visitor

20. Conclusion

The Visitor Management System successfully automates visitor entry management and approval workflows.
It improves security, reduces paperwork, and provides real-time visibility into visitor status.
This project demonstrates full-stack development skills using React, Spring Boot, and MySQL.